

Math Reference Cards — PNP–BV Solver Updates

For the May 2026 group meeting slides. Each card is a self-contained block intended to be screenshotted directly into a slide.

Baseline: 4-species PNP-BV (pre-rebuild)

Bulk Nernst–Planck (one equation per species i):

$$\frac{\partial c_i}{\partial t} = -\nabla \cdot J_i, \quad J_i = -D_i \left(\nabla c_i + \frac{z_i F}{RT} c_i \nabla \varphi \right)$$

Poisson:

$$-\varepsilon \nabla^2 \varphi = F \sum_i z_i c_i$$

Butler–Volmer flux BC at the electrode for reaction j :

$$r_j = k_{0,j} \left[c_O^{\text{surf}} \exp\left(-\alpha_j \frac{n_e F \eta_j}{RT}\right) - c_R^{\text{surf}} \exp\left((1 - \alpha_j) \frac{n_e F \eta_j}{RT}\right) \right]$$

Species (4-species formulation): O_2 , H_2O_2 , H^+ , ClO_4^- . Charges $z = (0, 0, +1, -1)$. Reactions



Convergence window: $V_{\text{RHE}} \in [-0.50, +0.10]$ V (cathodic edge only).

Change 1: 3-species + analytic Boltzmann counterion

Drop ClO_4^- as a dynamic Nernst–Planck unknown. Replace its contribution to Poisson by an analytic Boltzmann profile in local equilibrium with φ .

Modified Poisson source (nondimensional, ideal mode):

$$-\varepsilon \nabla^2 \varphi = F \sum_{i \in \{\text{O}_2, \text{H}_2\text{O}_2, \text{H}^+\}} z_i c_i - F c_{\text{ClO}_4}^{\text{bulk}} \exp(-z_{\text{ClO}_4} \varphi)$$

Why it works: a species in zero-flux equilibrium with the field satisfies $\nabla(\ln c_i + z_i \varphi) = 0$, i.e. $c_i \propto \exp(-z_i \varphi)$. For an inert supporting electrolyte this is exact in the chemically active region; the inert co-ion never needs to be discretized.

Convergence window after Change 1: cold-starts in $V_{\text{RHE}} \gtrsim -0.20$ V; warm-walk fills the full $V_{\text{RHE}} \in [-0.50, +0.10]$ V grid with sub-0.5% agreement to the 4-species reference (V24 validation, 8/8 PASS).

Change 2: log-concentration formulation

Switch primary unknowns from c_i to $\boxed{u_i = \ln c_i}$. Reconstruct $c_i = \exp(u_i)$ which is positive by construction.

Nernst–Planck flux in log-c form:

$$J_i = -D_i c_i (\nabla u_i + z_i \nabla \varphi) = -D_i e^{u_i} (\nabla u_i + z_i \nabla \varphi)$$

Backward-Euler weak form:

$$\int_{\Omega} \frac{e^{u_i} - e^{u_i^{\text{old}}}}{\Delta t} v \, d\Omega + \int_{\Omega} D_i e^{u_i} (\nabla u_i + z_i \nabla \varphi) \cdot \nabla v \, d\Omega = (\text{BV boundary terms})$$

Poisson source in log-c form:

$$\int_{\Omega} \varepsilon \nabla \varphi \cdot \nabla w \, d\Omega - \int_{\Omega} \rho_{\text{rhs}} \sum_i z_i e^{u_i} w \, d\Omega = 0$$

Why it helps: $c_i = e^{u_i} > 0$ by construction; the Debye-layer behavior $c_i \sim c_{\text{bulk}} \exp(-z_i \varphi / V_T)$ becomes approximately linear in u_i ; SNES iterates can no longer overshoot to negative concentrations.

Change 3: log-rate Butler–Volmer

Inside the BV boundary residual, build the rate *additively in log space* and exponentiate once at the end.

Cathodic branch of reaction j :

$$\begin{aligned} \log r_j^{(\text{cat})} &= \ln k_{0,j} + u_{\text{cat}(j)} + \sum_f \nu_f (u_{\text{sp}(f)} - \ln c_{\text{ref},f}) - \alpha_j n_e \eta_j / V_T \\ r_j^{(\text{cat})} &= \exp(\log r_j^{(\text{cat})}) \end{aligned}$$

Anodic branch:

$$\log r_j^{(\text{anod})} = \ln k_{0,j} + u_{\text{anod}(j)} + (1 - \alpha_j) n_e \eta_j / V_T, \quad r_j^{(\text{anod})} = \exp(\log r_j^{(\text{anod})})$$

What this fixes: the legacy form evaluates $c_{\text{surf},i} = \exp(\text{clamp}(u_i, \pm 30))$ before multiplying by the BV exponential. At anodic V_{RHE} , $u_{\text{H}_2\text{O}_2}$ becomes very negative; the clamp pins c_{surf} at a finite floor and the saturated cathodic exponential of R_2 produces a phantom sink. Adding u_i inside the exponent lets $\log r_j \rightarrow -\infty$ smoothly, which is the physically correct behavior.

Convergence window after Change 3: the production grid now reaches $V_{\text{RHE}} = +0.60$ V (was $\leq +0.20$ V).

Stern compact layer (Robin BC)

The legacy no-Stern model imposes $\varphi_s = \varphi_m = \varphi_{\text{applied}}$ at the electrode (Dirichlet). Real interfaces split the applied drop:

$$\varphi_m - \varphi_{\text{bulk}} = \Delta\varphi_{\text{Stern}} + \Delta\varphi_{\text{Diffuse}}$$

Compact-layer capacitance law:

$$\sigma = C_S (\varphi_m - \varphi_s)$$

Robin boundary contribution to the Poisson residual (sign matches `forms_logc.py`):

$$F_{\text{res}} = - \underbrace{\tilde{C}_S}_{\text{nondim}} (\varphi_{\text{applied}} - \varphi) w \, d\Gamma_{\text{electrode}}$$

Modified BV overpotential under Stern:

$$\eta_j = \varphi_m - \varphi_s - E_{\text{eq}}^{(j)}$$

Limiting behavior:

$C_S \rightarrow \infty$: $\varphi_m - \varphi_s \rightarrow 0$ (recovers no-Stern Dirichlet); C_S finite : φ_s floats; some drop in the compact layer.

Why it matters: at $V_{\text{RHE}} > +0.66$ V the Boltzmann co-ion source $c_{\text{ClO}_4}^{\text{bulk}} \exp(\varphi_s)$ becomes catastrophically large under no-Stern. Finite C_S shifts part of the drop into the compact layer, lowering φ_s in the PNP domain.

May 4 result: on the production stack (3sp + Bikerman counterion + log-c- μ_H) with the Bikerman-consistent IC, $C_S = 0.10 \text{ F m}^{-2}$ gives 15/15 clean convergence on $V_{\text{RHE}} \in [-0.5, +1.0]$ V. At $V_{\text{RHE}} = +0.65$ V the Stern layer absorbs ≈ 12.8 V of the applied potential, keeping ψ_D modest enough that proton supply does not underflow the BV cathodic terms.

Bikerman steric correction

Finite ion size penalizes packing. Define the local packing fraction θ using ion sizes a_i (nondimensional):

$$\theta = 1 - \sum_{i \in \text{dyn}} a_i c_i - \underbrace{a_b c_{\text{steric},b}}_{\text{counterion contribution}}$$

Steric chemical potential added to the species' driving force:

$$\mu_i^{\text{steric}} = -\ln \theta \implies J_i = -D_i c_i (\nabla u_i + z_i \nabla \varphi + \nabla \mu_i^{\text{steric}})$$

Pre-steric (ideal) Boltzmann counterion — the legacy form already in use under Change 1 (zero-flux equilibrium in a point-ion PNP, no finite-size penalty on the counterion itself):

$$c_b(x) = c_b^{\text{bulk}} \exp(-z_b \varphi(x))$$

Contribution to the Poisson residual (`boltzmann.py:add_boltzmann_counterion_residual`):

$$F_{\text{res}} -= \rho_{\text{rhs}} z_b c_b^{\text{bulk}} \exp(-z_b \varphi) w d\Omega$$

This form is *unbounded* as $\varphi \rightarrow -\infty$ (for $z_b > 0$) or $\varphi \rightarrow +\infty$ (for $z_b < 0$) — the source of the $V_{\text{RHE}} > +0.66$ V Poisson blow-up before Stern was added.

Steric-aware analytic Boltzmann counterion (the Bikerman closure derived in `steric_analytic_clo4_reduction_handoff.md`):

$$c_b(x) = c_b^{\text{bulk}} \frac{(1 - \sum_{j \in \text{dyn}} a_j c_j(x)) \exp(-z_b \varphi(x))}{\theta_b + a_b c_b^{\text{bulk}} \exp(-z_b \varphi(x))}$$

where $\theta_b = 1 - \sum_j a_j c_j^{\text{bulk}} - a_b c_b^{\text{bulk}}$ is the bulk packing fraction (must be positive). The denominator caps c_b at $1/a_b$ so the modified Poisson source stays bounded everywhere.

Limiting behavior: $a \rightarrow 0$ recovers the ideal Boltzmann profile $c \propto \exp(-z \varphi)$ above.

Debye–Boltzmann analytical IC

Composite outer-electroneutral profile + inner Gouy–Chapman Debye layer. Picard-iterate the two-rate algebraic mass balance, then build φ and u_i on the production mesh.

Outer linear profiles (ambipolar diffusion gives the proton its effective $2D_H$ factor):

$$O_s = O_b - \frac{R_1}{D_O}, \quad P_s = P_b + \frac{R_1 - R_2}{D_P}, \quad H_o = H_b - \frac{R_1 + R_2}{D_H}$$

Diffuse-layer Boltzmann correction:

$$H_s = H_o \exp(-\psi_D), \quad \psi_D = \varphi_{\text{electrode}} - \varphi_o, \quad \varphi_o = \ln(H_o/c_-^{\text{bulk}})$$

2×2 algebraic BV system (one Picard step):

$$\begin{aligned} A_1 &= k_1(H_s/H_{\text{ref}})^2 \exp(-\alpha_1 n \eta_1), & B_1 &= k_1 \exp((1 - \alpha_1) n \eta_1), \\ A_2 &= k_2(H_s/H_{\text{ref}})^2 \exp(-\alpha_2 n \eta_2), \end{aligned}$$

$$\begin{bmatrix} 1 + A_1/D_O + B_1/D_P & -B_1/D_P \\ -A_2/D_P & 1 + A_2/D_P \end{bmatrix} \begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} A_1 O_b - B_1 P_b \\ A_2 P_b \end{bmatrix}$$

Gouy–Chapman composite for the potential:

$$\lambda_D = \sqrt{\varepsilon/(2H_o)}, \quad \psi(y) = 4 \operatorname{atanh}\left(\tanh(\psi_D/4) e^{-y/\lambda_D}\right)$$

$$\varphi_{\text{IC}}(y) = \ln(H_{\text{outer}}(y)/c_-^{\text{bulk}}) + \psi(y), \quad c_{H,\text{IC}}(y) = H_{\text{outer}}(y) e^{-\psi(y)}, \quad u_{i,\text{IC}} = \ln c_{i,\text{IC}}$$

Result: 1.7–6.3× fewer Newton iterations at $V_{\text{RHE}} \geq +0.30$ V; unblocks +0.60 V cold-start where the linear IC fails outright. Extended further by the Bikerman-consistent IC on the next card.

Bikerman-consistent IC (Option 2b)

Refines the Debye–Boltzmann IC for the steric (Bikerman) production stack. Two pieces:

(1) **Composite ψ profile** (Bazant–Kilic–Storey–Ajdari matched-asymptotic):

$$\nu = 2 a_{\text{Cl}} c_{\text{Cl}}^{\text{bulk}}, \quad \psi_{\text{sat}} = \ln(2/\nu), \quad \alpha = \sqrt{\frac{2}{\nu \lambda_D^2} \ln(1 + \nu(\cosh \psi_D - 1))}$$

$$y_{\text{match}} = \frac{|\psi_D| - \psi_{\text{sat}}}{\alpha}$$

$$\psi(y) = \begin{cases} \text{sgn}(\psi_D) (|\psi_D| - \alpha y) & 0 \leq y < y_{\text{match}} \\ \text{sgn}(\psi_D) \psi_{\text{sat}} \exp(-(y - y_{\text{match}})/\lambda_D) & y \geq y_{\text{match}} \end{cases}$$

Falls back to the tanh/Gouy–Chapman expression of the previous card when $|\psi_D| \leq \psi_{\text{sat}}$ (linear-Debye limit).

(2) **Multispecies activity correction γ :**

$$\gamma(y) = \frac{1}{1 + a_H H_{\text{outer}}(y) (e^{-\psi(y)} - 1) + a_{\text{Cl}} c_{\text{Cl}}^{\text{anchor}} (e^{+\psi(y)} - 1)}$$

(neutral species drop out automatically since $e^0 - 1 = 0$).

Seed every species on the saturated steric manifold:

$$\begin{aligned} u_{\text{O}_2}^{\text{IC}} &= \ln c_{\text{O}_2}^{\text{outer}} + \ln \gamma \\ u_{\text{H}_2\text{O}_2}^{\text{IC}} &= \ln c_{\text{H}_2\text{O}_2}^{\text{outer}} + \ln \gamma \\ u_H^{\text{IC}} &= \ln H_{\text{outer}} - \psi(y) + \ln \gamma \\ \varphi^{\text{IC}} &= \ln(H_{\text{outer}}/c_{\text{Cl}}^{\text{bulk}}) + \psi(y) \end{aligned}$$

μ_H **form:** same construction; ψ cancels exactly, $\mu_H^{\text{IC}} = 2 \ln H_{\text{outer}} - \ln c_{\text{Cl}}^{\text{bulk}} + \ln \gamma$.

Result (3sp+Bikerman+ μ_H , $N_y = 200$, clip 50, u-clamp 100): 15/15 cold/warm convergence on $V_{\text{RHE}} \in \{-0.5, \dots, +1.0\}$ V with Stern 0.10 F/m²; 14/15 without Stern. Cold ceiling +0.60 V (was +0.30 V), warm-walk reach +1.00 V (was +0.60 V).

Electrochemical-potential (μ_H) formulation

Replace the proton primary unknown u_H with the electrochemical potential

$$\boxed{\mu_H = u_H + z_H \varphi} \quad (z_H = +1)$$

so that $c_H = \exp(\mu_H - z_H \varphi)$ is reconstructed from the muh unknown plus the resolved field.

Nernst–Planck flux collapses (no separate drift term):

$$J_H = -D_H c_H \nabla \mu_H$$

Why it helps in the Debye layer: when H^+ is in local Boltzmann equilibrium, $\nabla \mu_H \approx 0$. The primary variable μ_H is therefore nearly flat in y even when u_H and φ each vary by tens of log-units across the Debye layer. Newton sees a smooth unknown in exactly the regime where u_H is most wild.

Cancellation under the Debye-Boltzmann IC:

$$\mu_H^{\text{IC}}(y) = u_H^{\text{IC}}(y) + \varphi^{\text{IC}}(y) = [\ln H_{\text{outer}} - \psi(y)] + [\ln(H_{\text{outer}}/c_-^{\text{bulk}}) + \psi(y)] = 2 \ln H_{\text{outer}} - \ln c_-^{\text{bulk}}$$

The diffuse-layer ψ cancels exactly — a clean correctness check.

Non-mu species (O_2 , H_2O_2 — neutral) are unchanged from the log-c stack.

Local pH observable (deck-aligned)

The Mangan/Ruggiero deck reports IrOx-sensed local pH at the disk surface. Cheap to derive from the existing surface H^+ field (no new species, no new BC):

$$\text{pH}_{\text{local}}(t) = -\log_{10}(c_{H,\text{surf}}(t) / 1000)$$

with $c_{H,\text{surf}}$ in mol m^{-3} (so the /1000 converts to mol L^{-1}). In the log-c formulation this is even cheaper:

$$\text{pH}_{\text{local}}(t) = -\frac{1}{\ln 10}(u_{H,\text{surf}}(t) - \ln 1000).$$

Use: report alongside CD/PC. Compare local-vs-bulk pH trajectory against the Ruggiero 2022 RRDE/IrOx data.

RRDE transport: Levich diffusion length

Tie the abstract reference length L_{ref} to a physical RRDE rotation rate via the Levich diffusion-layer thickness:

$$L_{\text{eff}} \approx 1.61 D_i^{1/3} \nu^{1/6} \omega^{-1/2}$$

where ω is the angular rotation rate (rad s^{-1}), ν is the kinematic viscosity of the electrolyte ($\text{m}^2 \text{s}^{-1}$), and D_i is the diffusivity of the rate-limiting species ($\text{m}^2 \text{s}^{-1}$).

Worked numbers for water at 25 °C, O₂-limited:

$$D_{\text{O}_2} = 1.9 \times 10^{-9} \text{ m}^2 \text{s}^{-1}, \quad \nu = 8.9 \times 10^{-7} \text{ m}^2 \text{s}^{-1}$$

ω corresponds to Ω [rpm] by $\omega = 2\pi\Omega/60$:

Ω [rpm]	ω [rad s ⁻¹]	L_{eff} [μm]
400	41.9	~ 86
1600	167.6	~ 43
2500	261.8	~ 35

Use: promote L_{ref} from a generic nondimensionalization scale to an *experimental* parameter so that rotation-rate sweeps in the multi-experiment FIM design correspond to a real measurable knob.

Where we are vs. deck experiment (closure list)

Aligned (physics-framework level): PNP, Bikerman steric, finite Stern, two-step BV (R_1/R_2), peroxide observable, three dynamic species + analytic counterion, log-c, log-rate.

Not yet aligned (experimental closure):

- **Electrolyte identity.** The Ruggiero study uses $\text{H}_2\text{SO}_4 + \text{MOH} + \text{M}_2\text{SO}_4$, $M \in \{\text{Li}, \text{Na}, \text{K}, \text{Cs}\}$. Our solver carries ClO_4^- as a *generic* inert surrogate. Need a sulfate-family balancing-anion path for deck-matched runs.
- **Cation specificity.** Deck story is ion-specific. Current solver has a generic steric size $a_{\text{default}} = 0.01$; needs cation-specific radii and an explicit (or analytic) cation species.
- **Local pH observable.** See card above.
- **RRDE transport.** See card above.
- **Ring channel + selectivity.** Post-processing on the existing peroxide-current observable.

Headline: voltage-window expansion

	Pre-rebuild (4-species)	Apr 27 stack (3sp+B+log-c+log-rate)	May 4 stack (+ Bikerman-IC + Stern)
Cold-start window	$[-0.50, +0.10]$	$\gtrsim -0.20$	$\gtrsim -0.20$
Full grid (warm-walk)	$[-0.50, +0.10]$	$[-0.50, +0.60]$	$[-0.50, +1.00]$
Demonstrated reach	+0.10 V	+0.60 V	+1.00 V (Bikerman-IC, Stern; 15/ +2.00 V (log-c+steric+debye_boltzmann,
R ₂ unclips at	—	+0.495 V	+0.495 V
Local FIM cond(F)	2.0×10^{11}	1.79×10^7	TBD
Weak eigenvector	$\approx \log k_{0,2}$	$\approx \log k_{0,1}$	TBD

Caveat — kinetic dead zone above $\sim +0.5$ V. Convergence reaches +2.0 V, but converged currents drop to machine epsilon (10^{-12} – 10^{-16} mA/cm²) once $c_{H,\text{surf}} \approx \exp(-\psi_D)$ underflows the cathodic BV terms. This is real physics, not a clip artefact: the proton factor $(c_H/c_{\text{ref}})^2$ (R₁) and $(c_H/c_{\text{ref}})^2$ (R₂) underflow at $\psi_D > 30$. Inverse-relevant signal remains in $V_{\text{RHE}} \in [-0.5, +0.5]$ V.

R₂ unclip threshold derivation. The implementation clips $\eta_{\text{scaled}} = (V - E_{\text{eq}})/V_T$ to ± 50 before the αn_e multiplication. R₂ becomes unclipped when $|V - E_{\text{eq}}^{(2)}| < 50 V_T$, i.e.

$$V > E_{\text{eq}}^{(2)} - 50 V_T = 1.78 - 50 (0.02569) = +0.495 \text{ V vs RHE.}$$